Student GPA

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1 Data set

1.1 Raw Data

Our data set contains data on 632 first-year students from Carnegie Mellon University, University of Washington, and Notre dame University. The relevant dataset attributes are race, gender, first-generation, Total sleep, midpoint sleep, Daytime Sleep, cumulative GPA, and Term GPA. The students' sleep data was collected via fit-bits.

```
Rows: 634
Columns: 15
$ subject_id            <int> 185, 158, 209, 102, 174, 184, 255, 265, 343, 137~
$ study                 <int> 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, ~
$ cohort                <chr> "lac1", "lac1", "lac1", "lac1", "lac1", "lac1", ~
$ demo_race             <int> 1, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
$ demo_gender           <int> 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 0, ~
$ demo_firstgen         <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, ~
$ bedtime_mssd          <dbl> 0.11672695, 0.14168084, 1.52928949, 0.13014845, ~
$ TotalSleepTime        <dbl> 432.2000, 391.9310, 344.3043, 392.6207, 423.4211~
$ midpoint_sleep        <dbl> 458.6600, 364.4655, 560.8913, 416.4828, 368.7632~
$ frac_nights_with_data <dbl> 0.8620690, 1.0000000, 0.7931034, 1.0000000, 0.65~
$ daytime_sleep         <dbl> 24.160000, 13.137931, 14.956522, 54.551724, 10.5~
$ cum_gpa               <dbl> 3.00, 3.66, 3.57, 3.61, 3.21, 3.20, 3.40, 3.86, ~
$ term_gpa              <dbl> 3.38, 2.60, 3.07, 3.56, 4.00, 3.36, 3.19, 3.28, ~
$ term_units            <int> 73, 64, 63, 61, 61, 60, 60, 60, 60, 59, 59, 58, ~
$ Zterm_units_ZofZ      <dbl> 4.0552949, 2.4825341, 2.3077829, 1.9582805, 1.95~
```

```
   subject_id          study          cohort            demo_race
 Min.   :    1.0   Min.   :1.000   Length:634         Min.   :0.000
 1st Qu.:  178.0   1st Qu.:2.000   Class :character   1st Qu.:1.000
 Median :  358.5   Median :3.000   Mode  :character   Median :1.000
 Mean   :13005.9   Mean   :3.181                      Mean   :0.812
 3rd Qu.:  592.8   3rd Qu.:4.000                      3rd Qu.:1.000
 Max.   :99978.0   Max.   :5.000                      Max.   :1.000
                                                      NA's   :1
  demo_gender     demo_firstgen     bedtime_mssd       TotalSleepTime
 Min.   :0.0000   Min.   :0.0000   Min.   : 0.004505   Min.   :194.8
 1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.: 0.074694   1st Qu.:366.9
 Median :1.0000   Median :0.0000   Median : 0.135007   Median :400.4
 Mean   :0.5832   Mean   :0.1667   Mean   : 0.451688   Mean   :397.3
 3rd Qu.:1.0000   3rd Qu.:0.0000   3rd Qu.: 0.291698   3rd Qu.:430.1
 Max.   :1.0000   Max.   :2.0000   Max.   :20.849225   Max.   :587.7
 NA's   :3        NA's   :4
 midpoint_sleep  frac_nights_with_data daytime_sleep        cum_gpa
```

```
 Min.   :247.1   Min.   :0.2143        Min.   :  2.269   Min.   :1.210
 1st Qu.:345.2   1st Qu.:0.8214        1st Qu.: 23.098   1st Qu.:3.232
 Median :388.2   Median :0.9322        Median : 34.982   Median :3.558
 Mean   :398.7   Mean   :0.8674        Mean   : 41.164   Mean   :3.466
 3rd Qu.:437.7   3rd Qu.:1.0000        3rd Qu.: 51.249   3rd Qu.:3.790
 Max.   :724.7   Max.   :1.0000        Max.   :292.304   Max.   :4.000

    term_gpa       term_units    Zterm_units_ZofZ
 Min.   :0.350   Min.   : 5.00   Min.   :-3.98252
 1st Qu.:3.233   1st Qu.:15.00   1st Qu.:-0.55104
 Median :3.556   Median :17.00   Median : 0.04121
 Mean   :3.450   Mean   :29.39   Mean   : 0.00000
 3rd Qu.:3.810   3rd Qu.:48.00   3rd Qu.: 0.56027
 Max.   :4.000   Max.   :73.00   Max.   : 4.05530
                 NA's   :147     NA's   :147
```

1.2 Tidy Data

This is our first tidy dataset.

```
Rows: 634
Columns: 7
$ id             <int> 185, 158, 209, 102, 174, 184, 255, 265, 343, 137, 279, ~
$ naptime        <dbl> 24.160000, 13.137931, 14.956522, 54.551724, 10.526316, ~
$ total_sleep    <dbl> 432.2000, 391.9310, 344.3043, 392.6207, 423.4211, 397.4~
$ midpoint_sleep <dbl> 458.6600, 364.4655, 560.8913, 416.4828, 368.7632, 353.5~
$ bedtime_mssd   <dbl> 0.11672695, 0.14168084, 1.52928949, 0.13014845, 0.13018~
$ cum_gpa        <dbl> 3.00, 3.66, 3.57, 3.61, 3.21, 3.20, 3.40, 3.86, 3.79, 3~
$ term_gpa       <dbl> 3.38, 2.60, 3.07, 3.56, 4.00, 3.36, 3.19, 3.28, 3.50, 2~

       id             naptime         total_sleep    midpoint_sleep
 Min.   :    1.0   Min.   :  2.269   Min.   :194.8   Min.   :247.1
 1st Qu.:  178.0   1st Qu.: 23.098   1st Qu.:366.9   1st Qu.:345.2
 Median :  358.5   Median : 34.982   Median :400.4   Median :388.2
 Mean   :13005.9   Mean   : 41.164   Mean   :397.3   Mean   :398.7
 3rd Qu.:  592.8   3rd Qu.: 51.249   3rd Qu.:430.1   3rd Qu.:437.7
 Max.   :99978.0   Max.   :292.304   Max.   :587.7   Max.   :724.7
  bedtime_mssd          cum_gpa         term_gpa
 Min.   : 0.004505   Min.   :1.210   Min.   :0.350
 1st Qu.: 0.074694   1st Qu.:3.232   1st Qu.:3.233
 Median : 0.135007   Median :3.558   Median :3.556
 Mean   : 0.451688   Mean   :3.466   Mean   :3.450
 3rd Qu.: 0.291698   3rd Qu.:3.790   3rd Qu.:3.810
 Max.   :20.849225   Max.   :4.000   Max.   :4.000
```

This is our second tidy data set.

```
Rows: 625
Columns: 10
$ id             <dbl> 185, 158, 209, 102, 174, 184, 255, 265, 343, 137, 279, ~
$ race           <fct> Not Underrepresented, Underrepresented, Not Underrepres~
$ gender         <fct> Female, Female, Female, Female, Female, Female, Female,~
$ firstgen       <fct> Not First-Gen, Not First-Gen, Not First-Gen, First-Gen,~
$ total_sleep    <dbl> 432.2000, 391.9310, 344.3043, 392.6207, 423.4211, 397.4~
$ university     <fct> Carnegie Mellon University SP 2017, Carnegie Mellon Uni~
$ cum_gpa        <dbl> 3.00, 3.66, 3.57, 3.61, 3.21, 3.20, 3.40, 3.86, 3.79, 3~
$ term_gpa       <dbl> 3.38, 2.60, 3.07, 3.56, 4.00, 3.36, 3.19, 3.28, 3.50, 2~
$ midpoint_sleep <dbl> 458.6600, 364.4655, 560.8913, 416.4828, 368.7632, 353.5~
$ naptime        <dbl> 24.160000, 13.137931, 14.956522, 54.551724, 10.526316, ~

       id                           race        gender             firstgen
 Min.   :    1   Underrepresented    :118   Male  :261   Not First-Gen:522
 1st Qu.:  178   Not Underrepresented:507   Female:364   First-Gen    :103
 Median :  360
 Mean   :12946
 3rd Qu.:  595
 Max.   :99978
  total_sleep                                 university     cum_gpa
 Min.   :194.8   Carnegie Mellon SP 2018           : 76   Min.   :1.210
 1st Qu.:366.8   University of Washington SP 2018  :137   1st Qu.:3.230
 Median :400.5   University of Washington SP 2019  :137   Median :3.560
 Mean   :397.4   Notre Dame University SP 2016     :145   Mean   :3.465
 3rd Qu.:430.1   Carnegie Mellon University SP 2017:130   3rd Qu.:3.790
 Max.   :587.7                                            Max.   :4.000
    term_gpa     midpoint_sleep     naptime
 Min.   :0.350   Min.   :247.1   Min.   :  2.269
 1st Qu.:3.233   1st Qu.:345.7   1st Qu.: 23.130
 Median :3.556   Median :387.1   Median : 35.083
 Mean   :3.450   Mean   :398.7   Mean   : 41.276
 3rd Qu.:3.810   3rd Qu.:437.7   3rd Qu.: 51.571
 Max.   :4.000   Max.   :724.7   Max.   :292.304
```

2 Questions

2.1 what is the relationship between student GPA and their sleep?

We want to measure Student's cumulative and term GPA against their total sleep, daytime sleep, and midpoint sleep to see if there are any correlations or trends between them.

2.2 what is the relationship between student GPA and their demographic?

Do first-gen students perform better, or vise versa? we want to know if it makes a difference or not, and what kind of difference it makes.

Does race or gender have a relationship with GPA? Is there a combination of the 2 that has a strong affect on GPA that could possibly be stronger than the sleep variables?

3 Student GPA and Sleep

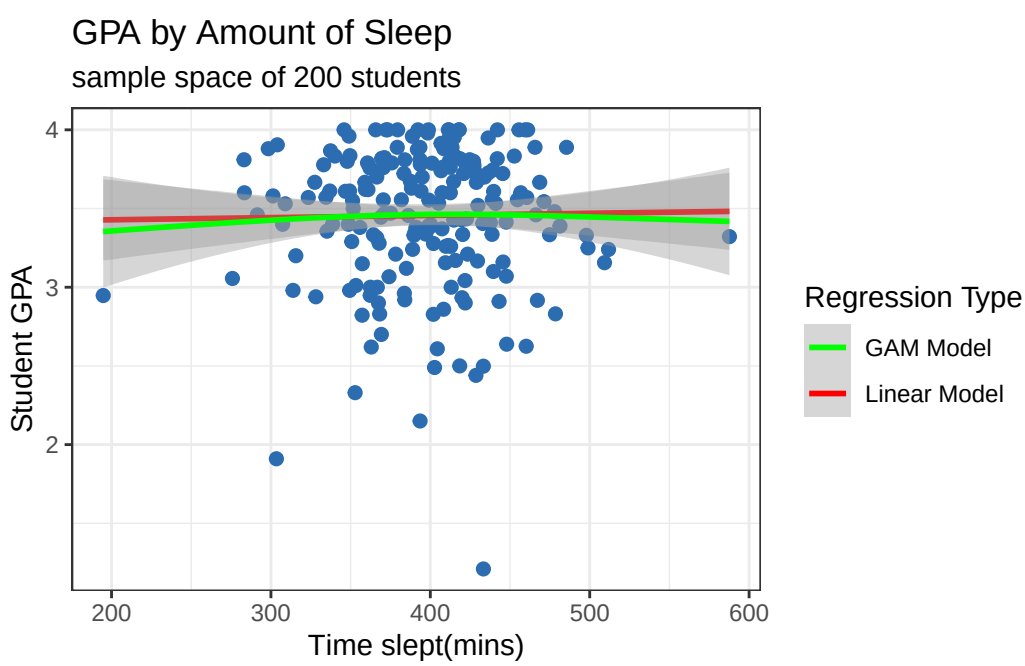
In this section, we explore the relationships between a Students GPA and several types of sleep.

3.1 Student GPA and Their Total Sleep

3.1.1 Graph

The following is a scatter plot with the y-axis as GPA and the x-axis as Total sleep. Note: Total sleep is the average sleep the student got during the semester, not the actual total amount of sleep. Each point on the graph is a single student that was randomly sampled from the data set.

Figure 1: scatterplot of the relationship between student GPA and their total sleep



3.1.2 Insights

When looking at the linear regression line, it shows a very weak fit and near zero correlation. Due to this, we can determine that there is no linear relationship/trend between a Students GPA and their total(average) sleep.

When looking at the GAM(non-linear) regression line, we can see nearly the same as the linear regression line, nearly no curve, weak fit. This tells us one of two things. That a students GPA and their average total sleep has no non-linear relation all, or the GAM Model is too simple to capture

the relationship, so a more flexible model may be needed. However, due to how similar the two regression lines are, it is unlikely that we need a more complex model.

For this, we can conclude that there is no meaningful relationship between a students GPA and their total sleep. Simply getting more sleep will not result in a student getting a better GPA, and getting less sleep not will hurt a students GPA.

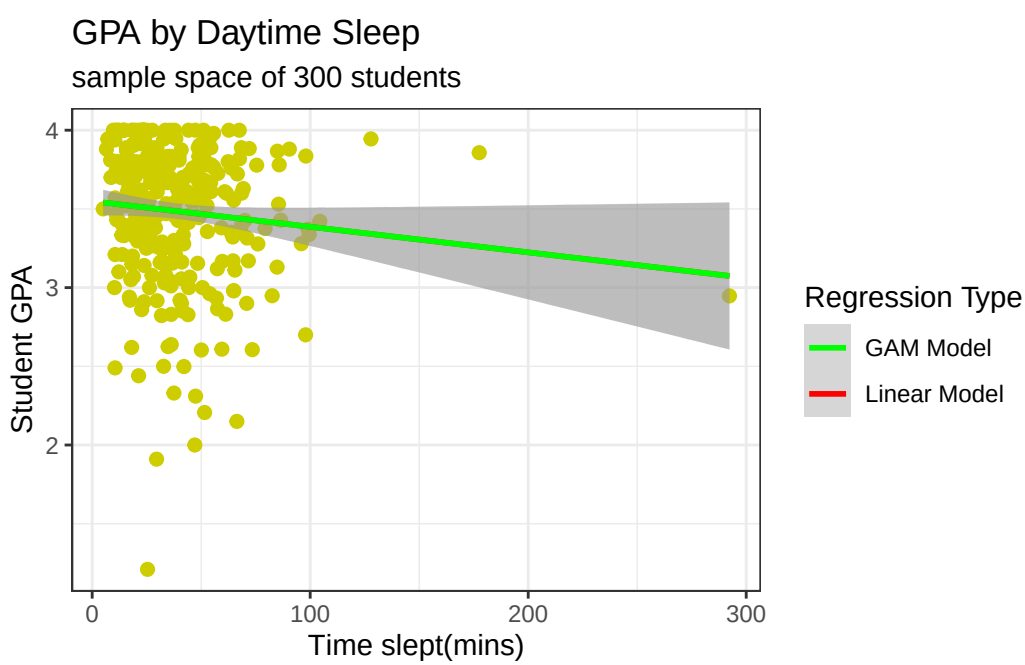
There are multiple factors to that could be affecting a students GPA, like how much they study, activities outside of school, or substance abuse. There is also the possibility that quality of sleep is a more deciding factor than the total sleep.

3.2 Student GPA and their Daytime-Sleep/Nap-time

This graph is like the last one, except we replaced total sleep with the students daytime sleep, or how much they napped.

3.2.1 Graph

Figure 2: scatterplot of the relationship between student GPA and how much they napped



3.2.2 Insights

As we can see, the linear and non-linear lines are identical. You can't even see the linear line. Both lines show a slight negative correlation. in addition, the model is confident on the left and unsure on the right.

What does this tell us?:

The identical regression lines tells that our linear model agrees with our non-linear model, and shows the relationship is simple, no curvature, no complexity, no nonlinear pattern. In other words, we don't need a more complex model.

However, the models are very unsure for students who take more naps. Unfortunately, this is due to there being very few long nappers in the data set.There is also a very weak fit, meaning the model doesn't accurately represent the data. Even if a trend was found, it wouldn't explain the data.

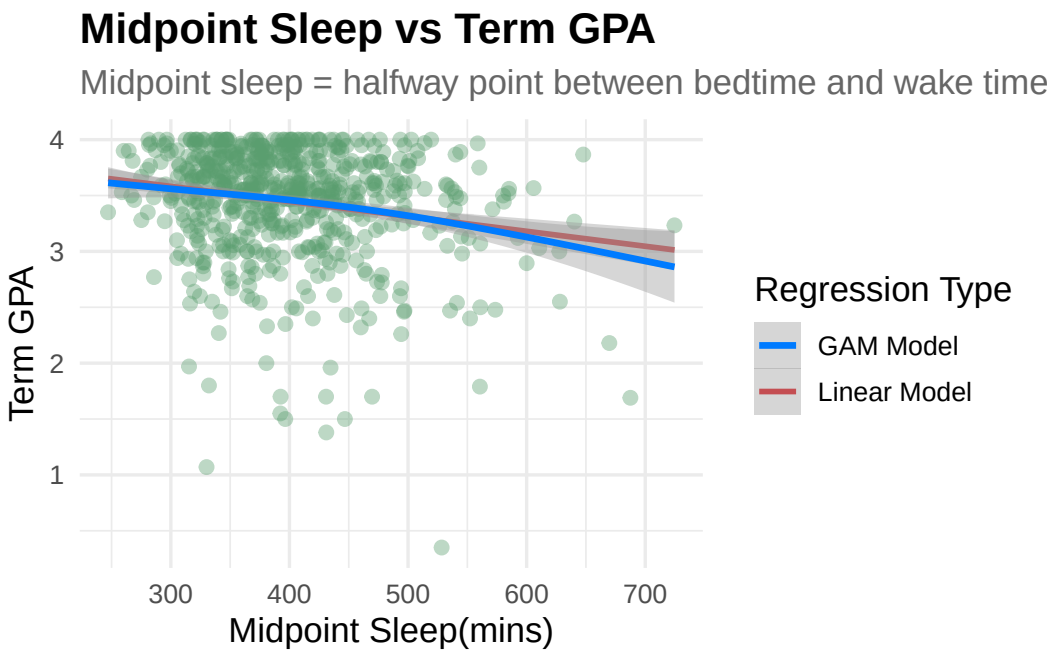
With all this together, we can conclude that there is no linear, non-linear, strong, or confident relationship. Therefore, there is no meaningful relationship between a students GPA and the time they napped.

3.3 Student GPA and Their Midpoint Sleep

Next, we graph the relationship between a students GPA and their midpoint sleep, midpoint sleep is the halfway point between when someone falls asleep and when they wake up. Midpoint sleep is relevant since it can tell us how early or late a student is sleeping. The higher value of midpoint sleep, the later the student went to bed.

3.3.1 Graph

Figure 3: scatterplot of the relationship between student GPA and late or early they sleep



3.3.2 Insights

Similar to Figure 1 and Figure 2, the linear and non-linear regression lines are almost identical. We can also see that the confidence of our model is very high. The overlap and high confidence tells us that it is highly unlikely for there to be a strong non-linear relationship. However, our graph has weak fit, evidence by the scatter.

What this really means is that there is a relationship between GPA and midpoint sleep, but it is very weak and not very useful.

3.4 Student GPA and Their Sleep Quality

Since we could not find meaningful relationships between student GPA, total sleep, daytime sleep, and midpoint sleep, we wanted to see if there we could find one the quality of sleep a student got.

Our raw data set does not have the quality of a students sleep, so we had to make a rough estimate using total sleep, daytime sleep, midpoint sleep, and bedtime_mssd(midpoint standard deviation).

$$\text{Sleep Quality} \approx \frac{\text{duration score} + \text{regularity score} + \text{timing score}}{3}$$

$$\text{duration score} = \frac{\text{nightsleep} - \text{nightmin}}{\text{nightmax} - \text{nightmin}}$$

$$\text{nightsleep} = \text{totalsleep} - \text{daytimesleep}$$

$$\text{timing score} = 1 - \frac{|\text{midpoint sleep} - \text{timing center}|}{\text{timing tol}}$$

3.4.1 Graph

3.4.2 Insights

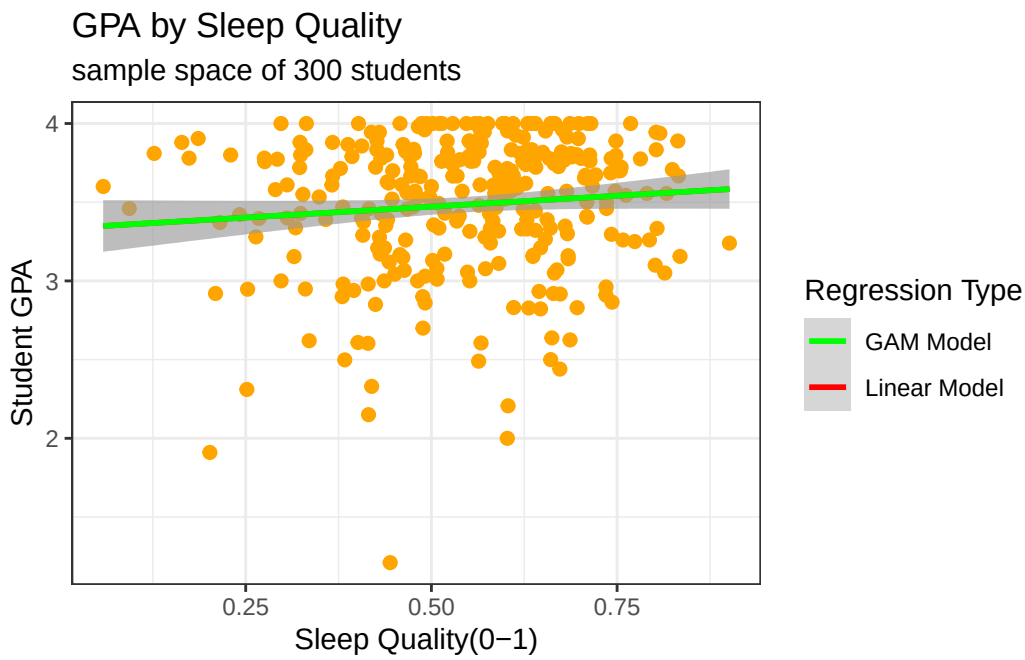
Unfortunately, it looks like not much changed. We can see from the graph that the model has low fit, and identical regression lines, virtually no slope, much like the last graphs. The only thing that improved was our models certainty. Like the previous graphs, the high certainty, low fit, and identical regression lines means that our data contains no meaningful pattern, and the model is very sure that there is no pattern.

Surprisingly, it seems that sleep, whether its quantity or quality, doesn't have that big of an impact on a students GPA, implying there are much stronger factors other than sleep.

4 Student GPA and demographics

In this section, we compare first generation students with non first generation students and represented with under-represented students by GPA. Also we, compare different combinations of the two, along with their correlations.

Figure 4: scatterplot of the relationship between student GPA and Sleep Quality



4.1 Graph

This graph has multiple views showing amount of sleep by each measure, term GPA, and demographic combination (e.g. Not under-represented / First-Gen). Along with the graph there is a correlation table.

```
# A tibble: 4 x 4
  race_firstgen                        corr_total corr_midpoint corr_naptime
  <fct>                                     <dbl>         <dbl>        <dbl>
1 Not Underrepresented / First-Gen          0.158       -0.0663     -0.00375
2 Underrepresented / First-Gen              0.132       -0.321      -0.105
3 Not Underrepresented / Not First-Gen      0.191       -0.222      -0.175
4 Underrepresented / Not First-Gen          0.293       -0.160      -0.0590
```

4.2 Correlation Table

4.3 Insights

Across all groups, students who slept more tended to have slightly higher term GPAs, but the effect was small. The strongest and most consistent pattern came from sleep timing: students with later midpoints (meaning they go to bed and wake up later) generally earned lower GPAs. This was especially true for Underrepresented First-Gen students, who showed the steepest drop in GPA as midpoint sleep shifted later. The scatterplots back this up; the midpoint panels all slope downward, and the decline is most noticeable in that group.

Figure 5: Sleep variables (total sleep, midpoint sleep, and naptime) plotted against term GPA, faceted by combined race x first-generation status.

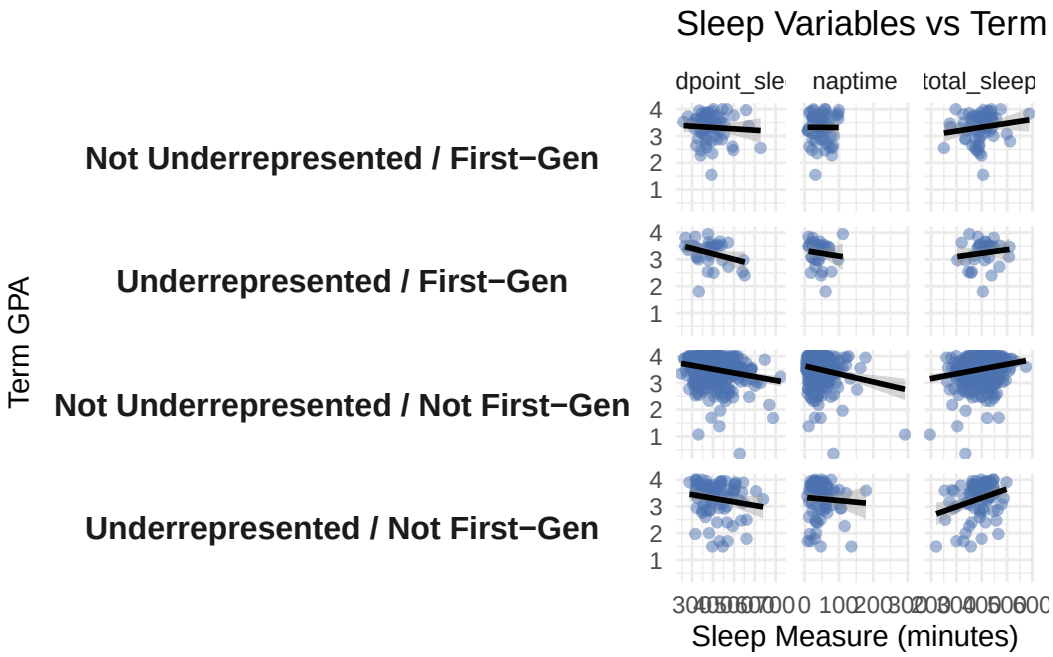


Table 1: Correlations between sleep variables and term GPA, computed within each combined race x first-generation demographic group.

Correlation of Sleep Variables with Term GPA

Grouped by Race × First-Generation Status

Race × First-Gen Group	Total Sleep	Midpoint Sleep	Naptime
Not Underrepresented / First-Gen	0.158	-0.066	-0.004
Underrepresented / First-Gen	0.132	-0.321	-0.105
Not Underrepresented / Not First-Gen	0.191	-0.222	-0.175
Underrepresented / Not First-Gen	0.293	-0.160	-0.059

Naptime didn't show much of a relationship with GPA. Most groups had correlations close to zero, and the trend lines in the naptime panels were basically flat. Overall, the results suggest that *when* students sleep matters more than *how much* they sleep, and that these patterns vary a across demographic groups. Some groups, particularly Underrepresented First-Gen students, seem more affected by later sleep timing, which could reflect differences in schedules, stress, or other outside factor.

5 Appendix

5.1 Code For Wrangling Data

```
Rows: 634
Columns: 15
$ subject_id            <int> 185, 158, 209, 102, 174, 184, 255, 265, 343, 137~
$ study                 <int> 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, ~
$ cohort                <chr> "lac1", "lac1", "lac1", "lac1", "lac1", "lac1", ~
$ demo_race             <int> 1, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
$ demo_gender           <int> 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 0, ~
$ demo_firstgen         <int> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, ~
$ bedtime_mssd          <dbl> 0.11672695, 0.14168084, 1.52928949, 0.13014845, ~
$ TotalSleepTime        <dbl> 432.2000, 391.9310, 344.3043, 392.6207, 423.4211~
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$ frac_nights_with_data <dbl> 0.8620690, 1.0000000, 0.7931034, 1.0000000, 0.65~
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$ term_gpa              <dbl> 3.38, 2.60, 3.07, 3.56, 4.00, 3.36, 3.19, 3.28, ~
$ term_units            <int> 73, 64, 63, 61, 61, 60, 60, 60, 60, 59, 59, 58, ~
$ Zterm_units_ZofZ      <dbl> 4.0552949, 2.4825341, 2.3077829, 1.9582805, 1.95~
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 Median :  358.5   Median :3.000   Mode  :character   Median :1.000
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 Max.   :99978.0   Max.   :5.000                      Max.   :1.000
                                                      NA's   :1
  demo_gender     demo_firstgen     bedtime_mssd       TotalSleepTime
 Min.   :0.0000   Min.   :0.0000   Min.   : 0.004505   Min.   :194.8
 1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.: 0.074694   1st Qu.:366.9
 Median :1.0000   Median :0.0000   Median : 0.135007   Median :400.4
 Mean   :0.5832   Mean   :0.1667   Mean   : 0.451688   Mean   :397.3
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 Max.   :1.0000   Max.   :2.0000   Max.   :20.849225   Max.   :587.7
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```

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    term_gpa       term_units    Zterm_units_ZofZ
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 Median :3.556   Median :17.00   Median : 0.04121
 Mean   :3.450   Mean   :29.39   Mean   : 0.00000
 3rd Qu.:3.810   3rd Qu.:48.00   3rd Qu.: 0.56027
 Max.   :4.000   Max.   :73.00   Max.   : 4.05530
                 NA's   :147     NA's   :147


Rows: 634
Columns: 7
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$ total_sleep    <dbl> 432.2000, 391.9310, 344.3043, 392.6207, 423.4211, 397.4~
$ midpoint_sleep <dbl> 458.6600, 364.4655, 560.8913, 416.4828, 368.7632, 353.5~
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$ term_gpa       <dbl> 3.38, 2.60, 3.07, 3.56, 4.00, 3.36, 3.19, 3.28, 3.50, 2~

       id             naptime         total_sleep    midpoint_sleep
 Min.   :    1.0   Min.   :  2.269   Min.   :194.8   Min.   :247.1
 1st Qu.:  178.0   1st Qu.: 23.098   1st Qu.:366.9   1st Qu.:345.2
 Median :  358.5   Median : 34.982   Median :400.4   Median :388.2
 Mean   :13005.9   Mean   : 41.164   Mean   :397.3   Mean   :398.7
 3rd Qu.:  592.8   3rd Qu.: 51.249   3rd Qu.:430.1   3rd Qu.:437.7
 Max.   :99978.0   Max.   :292.304   Max.   :587.7   Max.   :724.7
  bedtime_mssd          cum_gpa         term_gpa
 Min.   : 0.004505   Min.   :1.210   Min.   :0.350
 1st Qu.: 0.074694   1st Qu.:3.232   1st Qu.:3.233
 Median : 0.135007   Median :3.558   Median :3.556
 Mean   : 0.451688   Mean   :3.466   Mean   :3.450
 3rd Qu.: 0.291698   3rd Qu.:3.790   3rd Qu.:3.810
 Max.   :20.849225   Max.   :4.000   Max.   :4.000


Rows: 625
Columns: 10
$ id             <dbl> 185, 158, 209, 102, 174, 184, 255, 265, 343, 137, 279, ~
$ race           <fct> Not Underrepresented, Underrepresented, Not Underrepres~
$ gender         <fct> Female, Female, Female, Female, Female, Female, Female,~
$ firstgen       <fct> Not First-Gen, Not First-Gen, Not First-Gen, First-Gen,~
```

```
$ total_sleep    <dbl> 432.2000, 391.9310, 344.3043, 392.6207, 423.4211, 397.4~
$ university     <fct> Carnegie Mellon University SP 2017, Carnegie Mellon Uni~
$ cum_gpa        <dbl> 3.00, 3.66, 3.57, 3.61, 3.21, 3.20, 3.40, 3.86, 3.79, 3~
$ term_gpa       <dbl> 3.38, 2.60, 3.07, 3.56, 4.00, 3.36, 3.19, 3.28, 3.50, 2~
$ midpoint_sleep <dbl> 458.6600, 364.4655, 560.8913, 416.4828, 368.7632, 353.5~
$ naptime        <dbl> 24.160000, 13.137931, 14.956522, 54.551724, 10.526316, ~

       id                           race        gender             firstgen
 Min.   :    1   Underrepresented    :118   Male  :261   Not First-Gen:522
 1st Qu.:  178   Not Underrepresented:507   Female:364   First-Gen    :103
 Median :  360
 Mean   :12946
 3rd Qu.:  595
 Max.   :99978
  total_sleep                                 university     cum_gpa
 Min.   :194.8   Carnegie Mellon SP 2018           : 76   Min.   :1.210
 1st Qu.:366.8   University of Washington SP 2018  :137   1st Qu.:3.230
 Median :400.5   University of Washington SP 2019  :137   Median :3.560
 Mean   :397.4   Notre Dame University SP 2016     :145   Mean   :3.465
 3rd Qu.:430.1   Carnegie Mellon University SP 2017:130   3rd Qu.:3.790
 Max.   :587.7                                            Max.   :4.000
    term_gpa     midpoint_sleep     naptime
 Min.   :0.350   Min.   :247.1   Min.   :  2.269
 1st Qu.:3.233   1st Qu.:345.7   1st Qu.: 23.130
 Median :3.556   Median :387.1   Median : 35.083
 Mean   :3.450   Mean   :398.7   Mean   : 41.276
 3rd Qu.:3.810   3rd Qu.:437.7   3rd Qu.: 51.571
 Max.   :4.000   Max.   :724.7   Max.   :292.304
```

5.2 Code For Total Sleep Plot

```r
library(tidyverse)
library(ggplot2)
library(dplyr)
library(scales)

url <- "https://raw.githubusercontent.com/Stat184-Spring2026/Sec4_CP_Ben_Andrew_Mea/refs/heads/

sleep_clean <- read.csv(url)

set.seed(123)
sample_size <- 200 #change this value if you want a different sample size
sleep_clean |>
  slice_sample(n = sample_size, replace = FALSE) |>
  ggplot(
```

```
    mapping = aes(x = total_sleep, y = cum_gpa)
  ) + labs(
    title = "GPA by Amount of Sleep",
    subtitle = sprintf("sample space of %s students",sample_size),
    x = "Time slept(mins)",
    y = "Student GPA"
  ) + geom_point(size = 2, color = "#2C6DB2") +
  geom_smooth(aes(color = "Linear Model"),method = "lm") +
  geom_smooth(aes(color = "GAM Model"),method = "gam", formula = y ~ s(x)) +
  scale_color_manual(values = c("Linear Model" = "red", "GAM Model" = "green"), name = "Regress
  theme_bw()
```

5.3 Code For Daytime Sleep Plot

```
library(tidyverse)
library(ggplot2)
library(dplyr)
library(scales)

url <- "https://raw.githubusercontent.com/Stat184-Spring2026/Sec4_CP_Ben_Andrew_Mea/refs/heads/

sleep_clean <- read.csv(url)

set.seed(123)
sample_size <- 300 #change this value if you want a different sample size
sleep_clean |>
  slice_sample(n = sample_size, replace = FALSE) |>
  ggplot(
    mapping = aes(x = naptime, y = cum_gpa)
  ) + labs(
    title = "GPA by Daytime Sleep",
    subtitle = sprintf("sample space of %s students",sample_size),
    x = "Time slept(mins)",
    y = "Student GPA"
  ) + geom_point(size = 2, color = "yellow3") +
  geom_smooth(aes(color = "Linear Model"),method = "lm") +
  geom_smooth(aes(color = "GAM Model"),method = "gam", formula = y ~ s(x)) +
  scale_color_manual(values = c("Linear Model" = "red", "GAM Model" = "green"), name = "Regress
  theme_bw()
```

5.4 Code For Midpoint Sleep Plot

```r
library(tidyverse)
library(ggplot2)
library(dplyr)
library(scales)

ggplot(sleep_clean, aes(x = midpoint_sleep, y = term_gpa)) +
  geom_point(alpha = 0.4, color = "#5A9E6F", size = 2) +
  geom_smooth(aes(color = "Linear Model"),method = "lm", se = TRUE, linewidth = 1) +
  geom_smooth(aes(color = "GAM Model"),method = "gam", formula = y ~ s(x), se = TRUE) +
    scale_color_manual(values = c("Linear Model" = "#C44E52", "GAM Model" = "#007BFF"), name =
  labs(
    title    = "Midpoint Sleep vs Term GPA",
    subtitle = "Midpoint sleep = halfway point between bedtime and wake time(minutes after midn
    x        = "Midpoint Sleep(mins)",
    y        = "Term GPA",
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title    = element_text(face = "bold"),
    plot.subtitle = element_text(color = "gray40"),
    plot.caption  = element_text(color = "gray50")
  )
```

5.5 Code for Quality Sleep Figure

```r
library(tidyverse)
library(ggplot2)

sleep_quality_vec <- function(total_sleep,
                              daytime_sleep,
                              midpoint_sleep,
                              bedtime_mssd_days,
                              night_min = 180,    # 3h -> 0
                              night_max = 540,    # 9h -> 1
                              regularity_cut_min = 0,   # 0 min -> best
                              regularity_cut_max = 300, # 300 min -> worst (0)
                              timing_center = 400,      # minutes after midnight
                              timing_tol = 240) {       # minutes for timing tolerance (0->1 ma

  # convert mssd from days to minutes
  bedtime_mssd_min <- bedtime_mssd_days * 1440

  # nocturnal sleep
  night_sleep <- total_sleep - daytime_sleep
```

```r
  # duration score: linear 0->1 between night_min and night_max
  duration_score <- (night_sleep - night_min) / (night_max - night_min)
  duration_score <- pmin(1, pmax(0, duration_score))

  # regularity score: linear 1 -> 0 as bedtime_mssd_min goes from 0 -> regularity_cut_max
  regularity_score <- 1 - (bedtime_mssd_min - regularity_cut_min) / (regularity_cut_max - regul
  regularity_score <- pmin(1, pmax(0, regularity_score))

  # timing score: 1 at timing_center, declines linearly to 0 at timing_tol
  timing_score <- 1 - (abs(midpoint_sleep - timing_center) / timing_tol)
  timing_score <- pmin(1, pmax(0, timing_score))

  # combine (simple average)
  sleep_quality <- (duration_score + regularity_score + timing_score) / 3

  return(sleep_quality)
}

url <- "https://raw.githubusercontent.com/Stat184-Spring2026/Sec4_CP_Ben_Andrew_Mea/refs/heads/
sleep_clean <- read.csv(url)

# Compute sleep_quality column
sleep_clean <- sleep_clean %>%
  mutate(
    sleep_quality = sleep_quality_vec(
      total_sleep = total_sleep,
      daytime_sleep = naptime,
      midpoint_sleep = midpoint_sleep,
      bedtime_mssd_days = bedtime_mssd,
      night_min = 180,
      night_max = 540,
      regularity_cut_max = 300,
      timing_center = median(midpoint_sleep, na.rm = TRUE),
      timing_tol = 240
    )
  )

set.seed(123)
sample_size <- 300 #change this value if you want a different sample size
sleep_clean |>
  slice_sample(n = sample_size, replace = FALSE) |>
  ggplot(
    mapping = aes(x = sleep_quality, y = cum_gpa)
  ) + labs(
    title = "GPA by Sleep Quality",
    subtitle = sprintf("sample space of %s students",sample_size),
    x = "Sleep Quality(0-1)",
```

```
    y = "Student GPA"
  ) + geom_point(size = 2, color = "orange") +
  geom_smooth(aes(color = "Linear Model"),method = "lm") +
  geom_smooth(aes(color = "GAM Model"),method = "gam", formula = y ~ s(x)) +
  scale_color_manual(values = c("Linear Model" = "red", "GAM Model" = "green"), name = "Regress
  theme_bw()
```

5.6 Code For Demographics Plot

```
library(tidyverse)


url <- "https://raw.githubusercontent.com/Stat184-Spring2026/Sec4_CP_Ben_Andrew_Mea/refs/heads/

sleep_clean2 <- read.csv(url)

# Combined Demographics Table

sleep_clean2 <- sleep_clean2 %>%
  mutate(race_firstgen = interaction(race, firstgen, sep = " / "))


# Computing Correlations
sleep_clean2 %>%
  group_by(race_firstgen) %>%
  summarise(
    corr_total = cor(total_sleep, term_gpa, use = "complete.obs"),
    corr_midpoint = cor(midpoint_sleep, term_gpa, use = "complete.obs"),
    corr_naptime = cor(naptime, term_gpa, use = "complete.obs")
  )

# One Plot per Demographic group

sleep_clean2 %>%
  select(term_gpa, total_sleep, midpoint_sleep, naptime, race_firstgen) %>%
  pivot_longer(
    cols = c(total_sleep, midpoint_sleep, naptime),
    names_to = "sleep_variable",
    values_to = "sleep_value"
  ) %>%
  ggplot(aes(x = sleep_value, y = term_gpa)) +
  geom_point(alpha = 0.5, color = "#4C72B0") +
  geom_smooth(method = "lm", se = TRUE, color = "black") +
  facet_grid(race_firstgen ~ sleep_variable, scales = "free_x") +
  labs(
```

```
    title = "Sleep Variables vs Term GPA by Race × First-Gen Status",
    x = "Sleep Measure (minutes)",
    y = "Term GPA"
  ) +
  theme_minimal() +
    facet_grid(race_firstgen ~ sleep_variable, scales = "free_x", switch = "y") +
     theme(
       strip.text.y.left = element_text(angle = 0, size = 12, face = "bold"),
       strip.placement = "outside"
  )
```

5.7 Code for Correlation Table

```
library(tidyverse)
library(gt)

# Load data
url <- "https://raw.githubusercontent.com/Stat184-Spring2026/Sec4_CP_Ben_Andrew_Mea/refs/heads/
sleep_clean2 <- read.csv(url)

# Create combined demographic variable
sleep_clean2 <- sleep_clean2 %>%
  mutate(race_firstgen = interaction(race, firstgen, sep = " / "))

# Compute correlations
corr_table <- sleep_clean2 %>%
  group_by(race_firstgen) %>%
  summarise(
    corr_total_sleep = cor(total_sleep, term_gpa, use = "complete.obs"),
    corr_midpoint_sleep = cor(midpoint_sleep, term_gpa, use = "complete.obs"),
    corr_naptime = cor(naptime, term_gpa, use = "complete.obs")
  ) %>%
  mutate(across(starts_with("corr_"), round, 3))  # round to 3 decimals

# Display as a formatted table
corr_table %>%
  gt() %>%
  tab_header(
    title = "Correlation of Sleep Variables with Term GPA",
    subtitle = "Grouped by Race × First-Generation Status"
  ) %>%
  cols_label(
    race_firstgen = "Race × First-Gen Group",
    corr_total_sleep = "Total Sleep",
    corr_midpoint_sleep = "Midpoint Sleep",
```

```
    corr_naptime = "Naptime"
  )
```